

DATABASE MANAGEMENT SYSTEMS

II B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS09	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To understand the basic concepts and the applications of database systems.
2. To master the basics of SQL and construct queries using SQL.
3. Topics include data models, database design, relational model, relational algebra, transaction control, concurrency control, storage structures and access techniques.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Gain knowledge of fundamentals of DBMS, database design and normal forms
2. Master the basics of SQL for retrieval and management of data.
3. Be acquainted with the basics of transaction processing and concurrency control.
4. Familiarity with database storage structures and access techniques

UNIT - I	DATABASE SYSTEM APPLICATIONS & INTRODUCTION TO DATABASE DESIGN	CLASSES: 10
Database System Applications: A Historical Perspective, File Systems versus a DBMS, the Data Model, Levels of Abstraction in a DBMS, Data Independence, Structure of a DBMS Introduction to Database Design: Database Design and ER Diagrams, Entities, Attributes, and Entity Sets, Relationships and Relationship Sets, Additional Features of the ER Model, Conceptual Design With the ER Model		
UNIT - II	INTRODUCTION TO THE RELATIONAL MODEL	CLASSES: 10
Introduction to the Relational Model: Integrity constraint over relations, enforcing integrity constraints, querying relational data, logical database design, introduction to views, destroying/altering tables and views. Relational Algebra, Tuple relational Calculus, Domain relational calculus		
UNIT - III	SQL & SCHEMA REFINEMENT	CLASSES: 10
SQL: QUERIES, CONSTRAINTS, TRIGGERS: form of basic SQL query, UNION, INTERSECT, and EXCEPT, Nested Queries, aggregation operators, NULL values, complex integrity constraints in SQL, triggers and active databases. Schema Refinement: Problems caused by redundancy, decompositions, problems related to decomposition, reasoning about functional dependencies, First, Second, Third normal forms, BCNF, lossless join decomposition, multivalued dependencies, Fourth normal form, Fifth normal form.		
UNIT - IV	TRANSACTION CONCEPT	CLASSES: 10
Transaction Concept, Transaction State, Implementation of Atomicity and Durability, Concurrent Executions, Serializability, Recoverability, Implementation of Isolation, Testing for serializability, Lock Based Protocols, Timestamp Based Protocols, Validation- Based Protocols, Multiple Granularity, Recovery and Atomicity, Log-Based Recovery, Recovery with Concurrent Transactions.		

UNIT - V	DATA ON EXTERNAL STORAGE	CLASSES: 10
Data on External Storage, File Organization and Indexing, Cluster Indexes, Primary and Secondary Indexes, Index data Structures, Hash Based Indexing, Tree based Indexing, Comparison of File Organizations, Indexes- Intuitions for tree Indexes, Indexed Sequential Access Methods (ISAM), B+ Trees: A Dynamic Index Structure		
TEXT BOOKS		
1. Database System Concepts, Silberschatz, Korth, McGraw hill, V edition.3rd Edition 2. Database Management Systems, Raghurama Krishnan, Johannes Gehrke, Tata Mc Graw Hill		
REFERENCE BOOKS		
1. Database Systems design, Implementation, and Management, Peter Rob & Carlos Coronel 7th Edition. 2. Fundamentals of Database Systems, Elmasri Navrate, Pearson Education 3. Introduction to Database Systems, C. J. Date, Pearson Education 4. Oracle for Professionals, The X Team, S.Shah and V. Shah, SPD. 5. Database Systems Using Oracle: A Simplified guide to SQL and PL/SQL, Shah, PHI. 6. Fundamentals of Database Management Systems, M. L. Gillenson, Wiley Student Edition		